

Automated OSINT Profiling and Synthetic Social Engineering: An LLM-Based Framework for Personalized Risk Assessment

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Abstract

This thesis explores the intersection of Large Language Models (LLMs) and Open Source Intelligence (OSINT) to automate the reconnaissance phase of social engineering. Current Cyber Threat Intelligence (CTI) often neglects the “Human Attack Surface,” focusing instead on technical vulnerabilities. This research proposes a framework that aggregates disparate public data points to build a comprehensive digital footprint of an individual. By feeding this unstructured data into an LLM-driven analysis engine, the system identifies specific psychological levers and life-context triggers to generate highly personalized phishing vectors. The goal is to provide a scalable method for organizations to quantify individual exposure and simulate sophisticated social engineering threats before they are exploited by real-world adversaries.

1 The Problem

Social engineering remains the primary entry point for major breaches, yet its defense is rarely personalized:

1. **The Data Fragmentation:** A target’s digital footprint is scattered across professional networks, social media, and leaked databases. Manually correlating these for risk assessment is non-scalable.
2. **The Reconnaissance Advantage:** Adversaries spend weeks profiling high-value targets. Organizations lack an automated tool to mirror this "Attacker’s View" and predict which employees are most susceptible based on their current public activity.
3. **Generic Awareness Training:** Current anti-phishing training uses generic templates, failing to simulate the hyper-personalized lures enabled by modern generative AI.

2 The Proposed Solution: Synthetic Social Engineering

I propose a modular pipeline that automates the transition from raw OSINT data to actionable attack modeling.

Module I: Identity Resolution & Data Aggregation

The system utilizes a suite of OSINT techniques to scrape and normalize data from multiple sources:

- **Professional/Technical Footprint:** LinkedIn (roles, projects) and GitHub (coding habits, repo activity).

- **Social/Personal Footprint:** Instagram, X (Twitter), and Reddit to extract interests, sentiment, and location metadata.
- **Exposure Checks:** Cross-referencing handles and emails against known credential leak databases.

Module II: LLM-Driven Vulnerability Analysis

The aggregated data is processed by an LLM (e.g., Llama-3-8B or GPT-4o) to synthesize a "Narrative Profile."

- **Contextual Analysis:** Identifying "Life Events" (job changes, conferences, travels) that serve as ideal timing for a lure.
- **Psychological Profiling:** Mapping the target's public persona to Cialdini's principles of influence (e.g., Authority, Scarcity, or Social Proof).

Module III: Personalized Lure Generation

The final output is a set of synthetic attack vectors. The LLM generates hyper-personalized phishing emails or SMS scripts that match the target's professional tone and current context, demonstrating the potential for automated, high-fidelity social engineering.

3 Implementation Strategy

- **OSINT Tools:** Integration of tools like *Maigret*, *Social-Analyzer*, and custom Python scrapers for API-based data collection.
- **The Triage Layer:** A Retrieval-Augmented Generation (RAG) approach to feed the target's specific footprint into the LLM context window without exceeding token limits.
- **Ethical Constraints:** Research will be conducted using mock profiles or opt-in targets to ensure compliance with GDPR and ethical research standards at Aarhus University.

4 Project Plan

Phase	Timeline	Key Deliverables
1. Research LLM-based Social Engineering.	Weeks 1–4	Literature review on OSINT
2. OSINT Engine	Weeks 5–9	Functional multi-source scraper and data normalizer.
3. Profiling Logic	Weeks 10–13	LLM prompt engineering for psychological profiling.
4. Attack Modeling	Weeks 14–16	Generation of synthetic lures and risk scoring.
5. Writing	Weeks 17–19	Final thesis submission (June 4, 2026).